Owen Swartrout

Bandpass Sallen Key Lab Report

2/7/2025

Circuits II

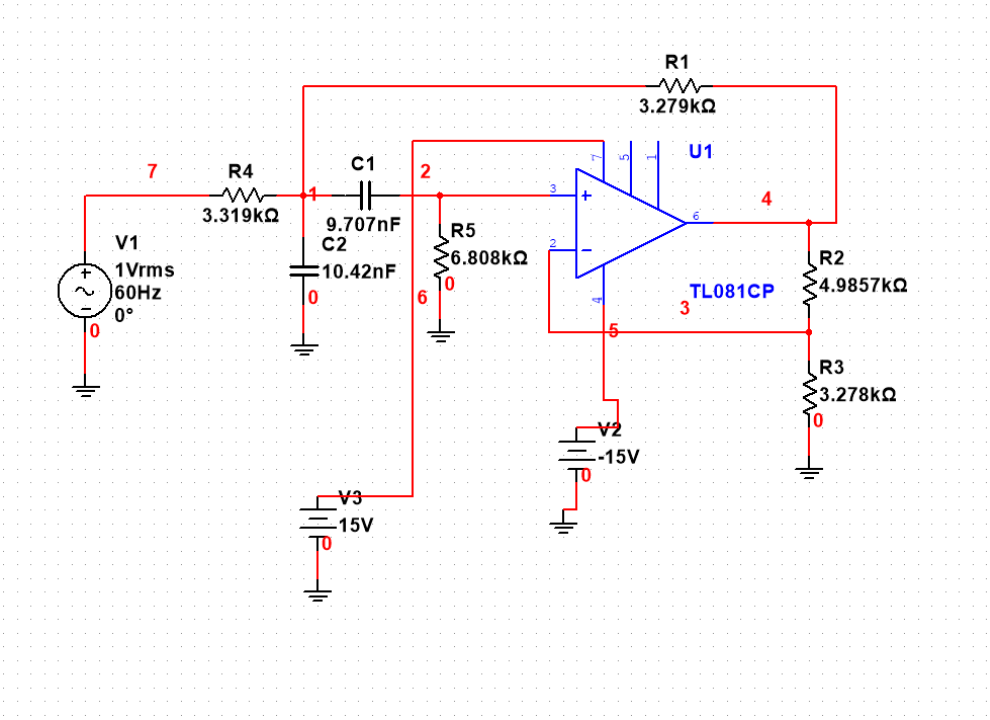


Figure 1: Circuit Diagram

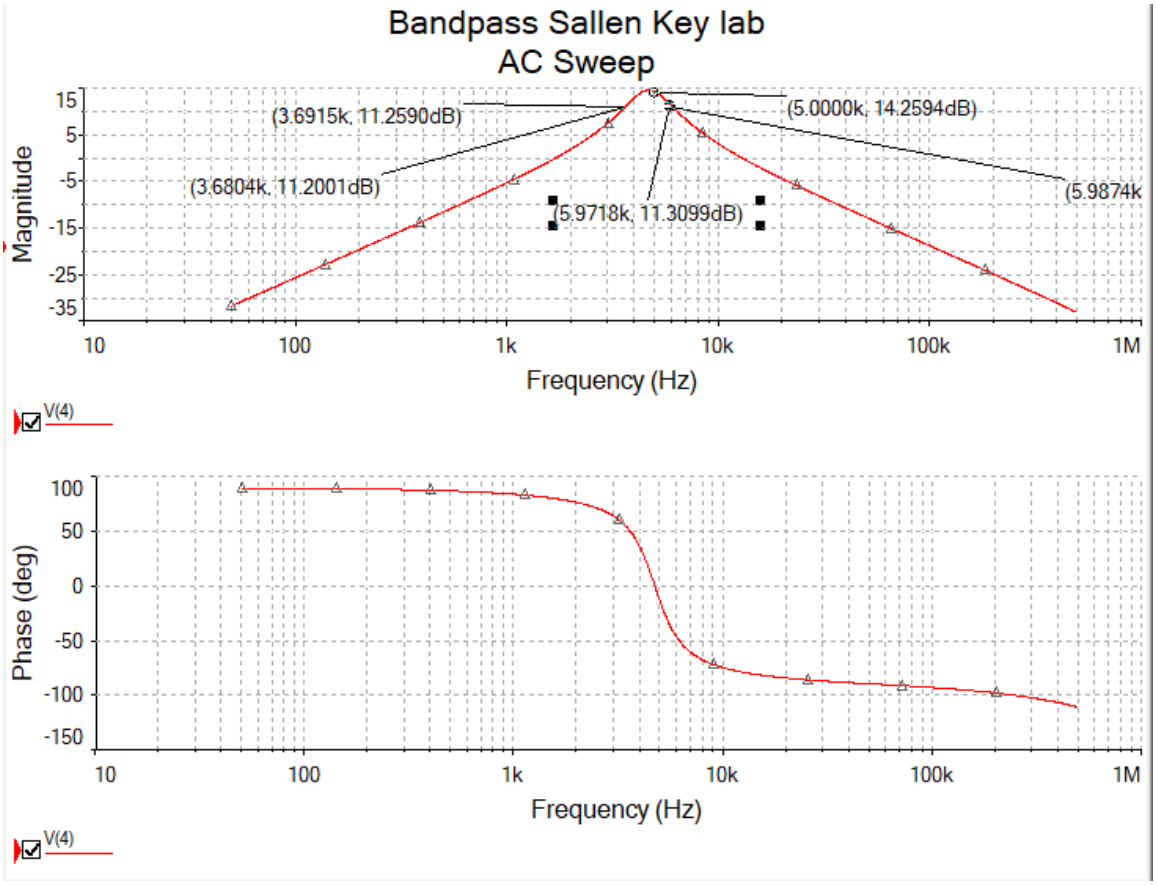


Figure 2: Multisim Simulation and Data

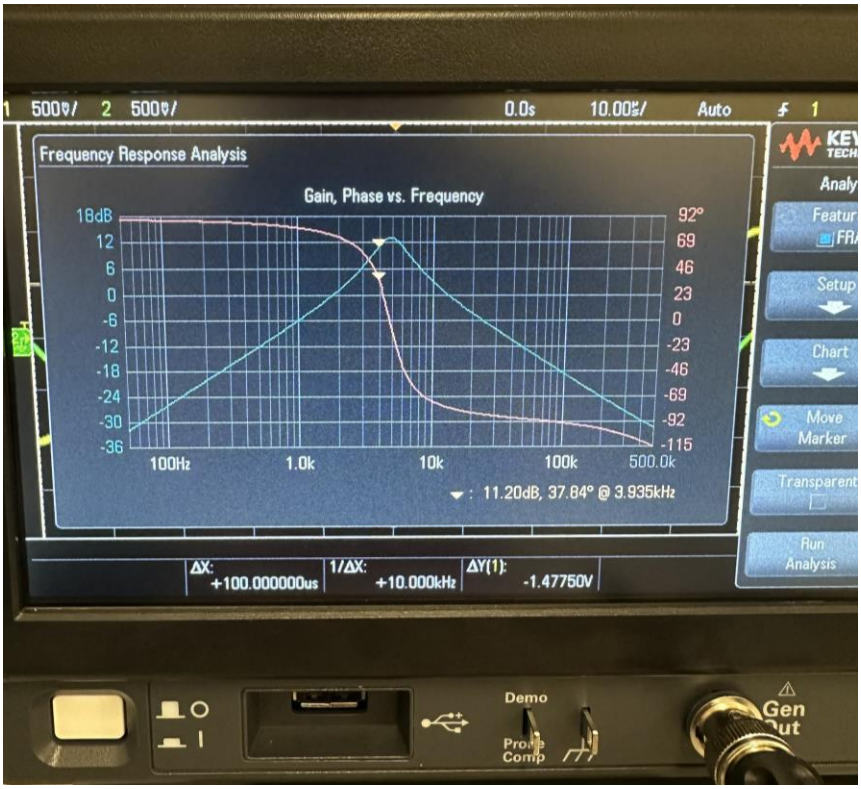


Figure 3: Scope Measurement at –3dB (to the left of the peak)

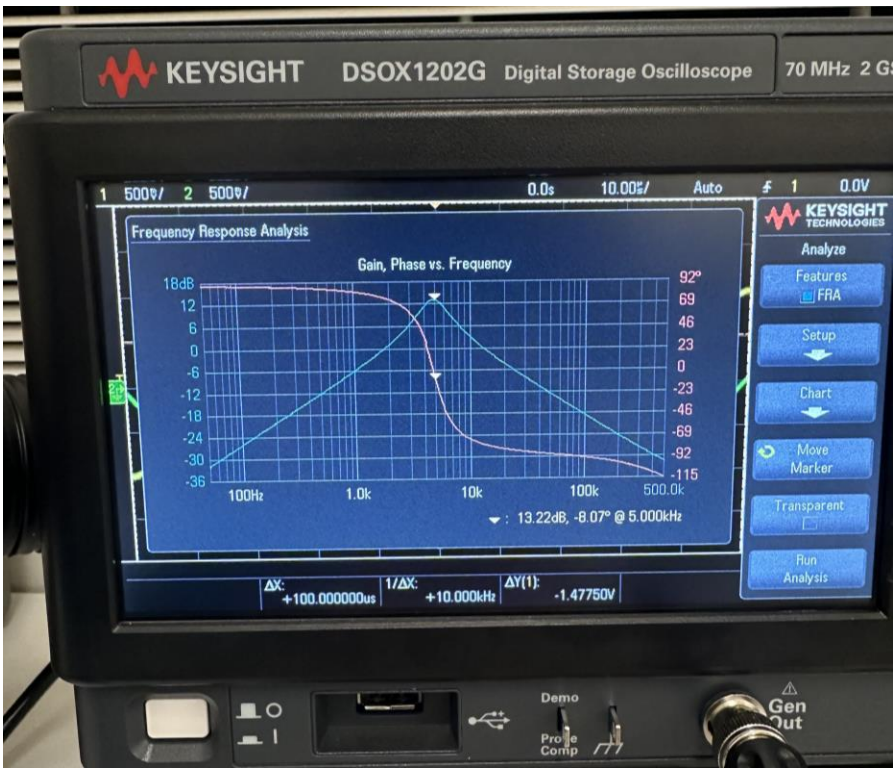


Figure 4: Scope Measurement at fnot (5kHz)

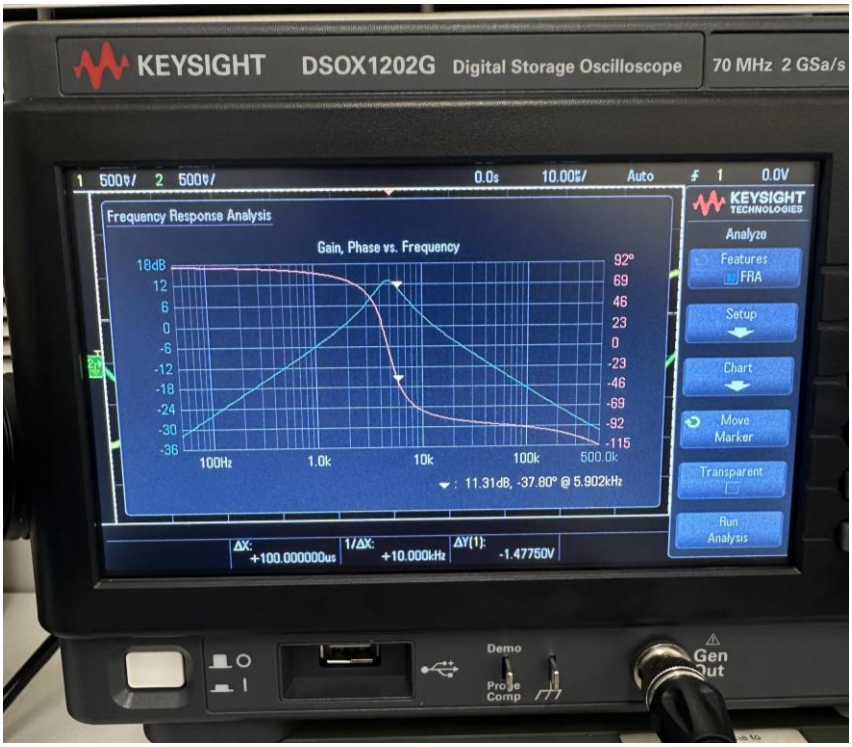


Figure 5: Scope Measurement at –3dB (to the right of the peak)

	Simulated Value	**Scope Value**	**Percent Error**
Gain at Peak (in dB)	14.26 dB	13.22dB	-7.29%
Frequency in Hz at –3dB left	3.69 kHz	3.935 kHz	6.64%
Frequency in Hz at –3dB right	5.987 kHz	5.902 kHz	-1.42%
Bandwidth	2.297 kHz	1.967 kHz	-14.37%

Analysis:

Overall, we have one outlier, the bandwidth. It is above the acceptable percentage error of 10%. This could be due to the capacitors on the circuit not fully functioning. Overall, the rest of the data is acceptable and looks good.